Containing Linux Instances with OpenVZ

Understanding the OpenVZ way of virtualisation and getting started with it.

Virtualisation is going mainstream, with many predicting that it will expand rapidly in the next few years. Virtualisation is a term that can refer to many different techniques. Most often, it is just software that presents a virtual hardware on which other software can run. Virtualisation is also done at a hardware level, like in the IBM mainframes or in the latest CPUs that feature the VT or SVM technologies from Intel and AMD, respectively. Although a fully featured virtual machine can run unmodified operating systems, there are other techniques in use that can provide special virtual machines, which are nevertheless very useful.

Comparison of virtualisation software

Virtualisation software	Ability to run unmodified guests	Performance	Level
QEMU	Yes	Relatively slow	User level
Xen*	No	Native	Below OS and above hardware
KVM	Yes	Native, but devices are emulated	Hardware-supported virtualisation
UML	No	Near native	OS on OS

Table 1

**Xen can make use of VT/SVM technology to run unmodified operating systems. In this case, it looks exactly like KVM.*

Performance and virtualisation

The x86 architecture is notorious for its virtualisation unfriendly nature. Explaining why this is the case requires a separate article on the subject. The only way to virtualise x86 hardware was to emulate it at the instruction level or to use methods like 'Binary Translation' and 'Binary Patching' at runtime. Well known software in this arena are QEMU, Vmware and the previously well-known Bochs. These programs emulate a full PC and can run unmodified operating systems.

The recent VT and SVM technologies provided by Intel and AMD, respectively, do away with the need to interpret/patch guest OS instruction streams. Since these recent CPUs provide hardware-level virtualisation, the virtualisation solution can trap into the host OS for any privileged operation that the guest is trying to execute.

Although running unmodified operating systems definitely has its advantages, there